

Experiment 5: Matrix Multiplication

Aim

To implement **Matrix Multiplication** using Python.

Algorithm

Step 1: Start.

Step 2: Read the number of rows and columns of the matrices.

Step 3: Read the elements of the first matrix.

Step 4: Read the elements of the second matrix.

Step 5: Multiply the corresponding rows and columns.

Step 6: Store the result in a new matrix.

Step 7: Display the resultant matrix.

Step 8: Stop.

Source Code (Python)

```
main.py
1 # Matrix Multiplication
2
3 r1 = int(input("Enter rows of Matrix A: "))
4 c1 = int(input("Enter columns of Matrix A: "))
5
6 r2 = int(input("Enter rows of Matrix B: "))
7 c2 = int(input("Enter columns of Matrix B: "))
8
9 if c1 != r2:
10     print("Matrix multiplication is not possible.")
11 else:
12     A = []
13     B = []
14
15     print("Enter Matrix A elements:")
16     for i in range(r1):
17         row = []
18         for j in range(c1):
19             row.append(int(input()))
20         A.append(row)
21
22     print("Enter Matrix B elements:")
23     for i in range(r2):
24         row = []
25         for j in range(c2):
26             row.append(int(input()))
27         B.append(row)
28
29     result = []
30
31     for i in range(r1):
32         row = []
33         for j in range(c2):
34             s = 0
35             for k in range(c1):
36                 s += A[i][k] * B[k][j]
37             row.append(s)
38         result.append(row)
39
40     print("Resultant Matrix:")
41     for row in result:
42         print(row)
```

Output

```
Output
Enter rows of Matrix A: 2
Enter columns of Matrix A: 2
Enter rows of Matrix B: 2
Enter columns of Matrix B: 2
Enter Matrix A elements:
1
2
3
4
Enter Matrix B elements:
5
6
7
8
Resultant Matrix:
[19, 22]
[43, 50]

=== Code Execution Successful ===
```

Time Complexity

- **Time Complexity:** $O(r_1 \times c_1 \times c_2)$
- **Space Complexity:** $O(r_1 \times c_2)$

Result

The **Matrix Multiplication** program was implemented successfully in Python. The program accepts two matrices, performs matrix multiplication, and displays the resultant matrix correctly.